

ZeroTrustAI: A Machine Learning-Driven Framework for Continuous Cybersecurity Risk Assessment

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Abstract—The rapid adoption of cloud technologies has led to a series of cybersecurity problems and exposed weaknesses in traditional security models. Traditional access-control methods rely on initial authentication and fixed trust, so they are less effective against behavioral changes, unexpected activity, and rule violations. This paper offers a machine learning-based Zero Trust architecture that combines machine learning, Retrieval-Augmented Generation, Explainable Artificial Intelligence, and continuous security monitoring. The architecture keeps on checking permissions using behavioral and contextual elements instead of granting constant trust to the previously verified user. A Random Forest algorithm is used for classification of attacks, and Isolation Forest for the detection of anomalies within the cloud activities. Gradient Boosting is used for assessing risks. Meanwhile, there is a policy analysis mechanism based on Retrieval-Augmented Generation technology for searching for needed policies from outside sources, while Explainable Artificial Intelligence enables to give understandable reasoning regarding the security decisions made with the help of Machine Learning technology. The Zero Trust Decision Engine separates incoming requests into ALLOW, VERIFY, RESTRICT, or DENY categories based on processed risk.

Index Terms—Zero Trust Architecture, Cybersecurity, Machine Learning, Anomaly Detection, Retrieval-Augmented Generation, Explainable Artificial Intelligence

I. INTRODUCTION

With the rapid development of cloud computing, web apps, remote access services, and distributed digital solutions, the cybersecurity environment today is completely different compared to previous years. Nowadays, organizations use various devices, networks, cloud systems, and other remote locations to access their resources, which increases the types of threats, such as compromised credentials, malicious use, insider attacks, illegal access, and violations of rules. Consequently,

conventional perimeter-based security models can no longer provide sufficient protection to digital resources in the modern world.

Zero Trust Architecture represents an alternative to current approaches by adopting continuous verification practices. Unlike current approaches that treat validated user/device as trusted, Zero Trust approaches every request according to its identity, device status, context of the request, sensitivity of the resource, and other security signals.

Zero Trust security can be enhanced by using machine learning to identify undesirable and unusual events using behavioral as well as security information. This framework implements the Random Forest technique for classifying risks of network attacks, the Isolation Forest technique for detecting anomalies occurring in the cloud, and the Gradient Boosting technique for assessing risks related to Zero Trust. They generate additional security signals that help evaluate various types of requests for access.

Another notable challenge faced is compliance with the security policy applied by the organization, which is normally introduced as a collection of documents defining the rules related to authentication, authorization, data access, privacy, and usage of the resources. Manual verification of the policy is ineffective; that is why the unique solution provided here is Retrieval-Augmented Generation (RAG) that is responsible for retrieving policies from external sources to offer them as evidence when analyzing violations within the policies.

Moreover, it is not easy for decision-makers to understand artificial intelligence. Explainable AI is applied to determine the most significant elements impacting the process of predicting threats and risks and to increase the level of transparency

of decisions made in the area of security. The suggested framework provides dominant means of distinguishing threats, assessing risk, verifying the policy, applying Explainable AI, and continuously monitoring the situation based on the Zero Trust principle.

The effectiveness of the framework is assessed in the controlled VirtualBox environment using normal and abnormal, suspicious, and policy-violating scenarios. The study aims to establish an adaptive, explainable, policy-aware, and context-sensitive access control system that responds to changes in the security environment.

II. LITERATURE REVIEW AND RESEARCH GAP

A. Literature Review

With the emergence of cyber threats, Zero Trust Architecture (ZTA) has become a valuable security solution for distributed, hybrid, and cloud scenarios. According to NIST, Zero Trust assumes that access to users and devices does not depend on their position in the network. In this regard, a user's identity, device, and context are analyzed to ensure that access is granted continuously [1], [2].

Studies have examined the synergistic use of Machine Learning (ML) technologies in Zero Trust-based security systems. Goshikonda and Pulipati developed a framework that combines ML algorithms, behavioral analytics, dynamic access control, and adaptive risk scoring [3]. Another example is SmartTrust presented by Lilhore et al. [4], in which CNN, LSTM, reinforcement learning, and transformers were implemented to detect threats in cloud services based on Zero Trust. Context-aware behavioral analysis was developed by Zhang and Wu [5], while Folino et al. created models for anomaly detection based on an ensemble approach [6]. Additionally, the solutions proposed by Manoharan et al. and Sarhan and Altwaijry prove the efficiency of supervised ML technology in the insider threat detection process [7], [8]. However, all the mentioned studies deal with anomaly detection rather than linking behavioral intelligence with continuous authorization in practice.

Dynamic access control has been examined as a method for adjusting permissions based on changing trust and risk conditions. Huang et al. introduced a trust-driven access control methodology for cloud-based environments [9]. On the other hand, Wang et al. developed a Zero Trust-based technique for managing access to cloud-based infrastructures [10]. Nevertheless, integration with respect to machine learning (ML)-based risk assessment is still not widely done.

Explainable AI (XAI) technology can help improve ML transparency in cybersecurity solutions. For example, Gaspar et al. and Sharma et al. used LIME and SHAP for making predictions regarding cyber intrusion detection [11], [12]. However, these works mostly focus on explanations for singular detection models and do not consider Zero Trust-based decisions.

Retrieval-Augmented Generation (RAG) enables security systems to achieve dynamic retrieval of recent organizational policies and security intelligence. For example, Lewis et

al. discussed RAG architecture [13]. Adediran et al. then applied RAG in conjunction with AWS CloudTrail and MITRE ATT&CK for cloud-based threat detection [14]. MITRE ATT&CK also provides valuable threat data that can be used for identifying effective techniques deployed by malicious users [15]–[17].

Commercial platforms currently offer a range of capabilities. Microsoft Defender for Identity can detect identity threats and analyze behavior. At the same time, Splunk Enterprise Security has risk-based security analytics, and CrowdStrike Falcon Identity Protection has behavior baselining and risk-based identity protection. The ZeroTrustAI framework does not intend to present the absence of ML, behavior analytics, or Zero Trust from these existing platforms; rather, its distinctiveness is to integrate these features among themselves within a common decision-making process.

B. Research Gap

Research has shown that zero-trust architecture, machine learning-based threat detection, combined access control, explainable AI, and retrieval-augmented generation work effectively as separate security mechanisms, but their potential is not exploited through their possible integration into a single, continuously working access-control framework. In particular, existing zero trust and machine learning methods mainly deal with threat detection and risk assessment tasks, while studies applying explainable AI focus on explanation of the results obtained by certain models. Similarly, retrieval-augmented generation technologies are primarily used for threat intelligence and knowledge retrieval tasks, instead of actual policy verification in the process of access decisions.

As a result, there is still a lack of a holistic Zero Trust framework for incorporating different types of ML security signals in the process of continuous risk assessment, the use of RAG for policy verification, as well as practical application of XAI for security reasoning. The ZeroTrustAI framework aims to fill this gap by bringing all of the aforementioned elements together. The unique feature of the ZeroTrustAI framework is that rather than relying on a single ML model for making access decisions, it relies on attack classification, anomaly detection, risk assessment, policy verification, and explainability for determining whether to ALLOW, VERIFY, RESTRICT, or DENY access.

III. PROPOSED FRAMEWORK

The suggested structure provides adaptive and contextual access control by employing Zero Trust Architecture, Machine Learning, Retrieval-Augmented Generation (RAG), Explainable Artificial Intelligence (XAI), and continuous security monitoring. Every request for access is evaluated using identity, device, network, cloud, behavioural, contextual, and legal information, instead of the limited criteria of authentication or location in the network.

A. Framework Architecture

The framework described includes gathering context, pre-processing data, analyzing security using machine learning,

carrying out risk evaluation, confirming policies in accordance with the RAG model, applying Explainable Artificial Intelligence (XAI), making the judgment, and ongoing monitoring of activities. The moment a person or device applies for access to a system resource, authentication and contextual information, including the status of the device, details of the applicant's request, network behavior, and cloud operations, are obtained. Authentication alone is not enough to confirm trust.

Security information received from different sources undergoes the stage of pre-processing and schema management, including cleaning of data, dealing with losses, transforming important data, converting numerical information, normalizing it where it is needed, and aligning schema. The information being processed is then received by the appropriate ML model.

B. ML-Based Security Analysis

Various ML models are applied for various security needs. Random Forest takes care of network threat classification, while Isolation Forest checks for abnormal activities in the cloud environment. Gradient Boosting ensures Zero Trust risk evaluation based on any relevant security and situational signals it can find. The results coming from the mentioned models are put together as complementary indicators, rather than being viewed as separate access decisions.

C. Risk-Based Zero Trust Assessment

The system constantly evaluates risks linked to an access request using further prediction technologies, some behavioral data, contextual characteristics, and policy compliance. Considering the overall security situation, low-risk and compliant requests obtain ALLOW, moderate or unsure ones are given VERIFY, suspicious acts receive RESTRICT, and high-risk requests or definite policy violations are assigned DENY. Thus, access decisions can be made according to the changing security situation.

D. RAG-Based Policy Verification

Retrieval-Augmented Generation (RAG) is employed to validate the compliance of events with the security rules of an organization. Security policy documentation that comprises provisions regarding authentication, authorization, security of information, privacy, as well as usage of resources is stored outside the organization. When an access request is received, relevant data is retrieved and input into the language model. After that, the request is measured against the retrieved information for compliance. Since the policy documents are maintained separately from the machine learning models, any change in the policies can be applied without updating the threat detection algorithms.

E. Explainable AI

The concept of Explainable Artificial Intelligence (XAI) makes it simpler to understand the comprehensive nature of decisions made with the help of ML in the field of security. Methods like SHAP and LIME effectively present the key parameters leading to the classification of a threat or prediction

of the risk. The conclusion received from the above-mentioned measures, together with information regarding relevant policies taken by RAG, will give the administrator the possibility to understand the reasons as well as the policies affecting the decision that has been made.

F. Unified Decision Engine and Continuous Monitoring

Zero Trust decision engine brings together the parts of authentication, contextual information, ML outputs, risk assessment, RAG policy outcomes, and XAI information to make the final access decision. No one ML model solely determines access. After access is granted, continuous monitoring of access continuously captures actions of users, devices, networks, and resources in the system. Any case of suspicious actions, anomalous cloud activities, or policy violations may lead to the reassessment of the request, and the state can be changed from ALLOW to VERIFY, RESTRICT, or DENY in the future.

The framework is analysed in a controlled environment employing VirtualBox in such a manner as to create typical, atypical, suspicious, and non-compliant conditions for simulating the comprehensive operation of the framework and demonstrating its capability to integrate detection of threats, risk assessment, verification of compliance with policies, explanation of the results, and implementation of constant monitoring into adaptive Zero Trust access control.

IV. SYSTEM IMPLEMENTATION AND EXPERIMENTAL SETUP

A. Data Collection and Preprocessing

The presented methodology integrates security data obtained from various sources in order to signify various components safeguarding an access request. For the industry tests, the CICIDS network intrusion dataset is used along with the application logs, authentication logs, SQLite database activity, fabricated cloud security events, as well as policies of the business organization. All of the provided sources represent information regarding the network, user identity, applications, cloud storage, as well as organizational policies needed for ZeroTrustAI-based decision-making.

As the investigated data is made up of different formats and feature representations, the preprocessing is performed prior to making a machine learning analysis. Thus, duplicates and wrongly obtained records are eliminated, whereas missing data is treated in accordance with the specifications of the concrete dataset.

Categorical variables have been changed into numerical values that can be used efficiently in machine learning models while numeric attributes are standardized when needed. Once data is processed, it gets structured according to the kind of security task solved by a particular model. Features associated with network problems are used in the Random Forest model, cloud and activity events are processed in Isolation Forest model, and variable features are applied to the Gradient Boosting risk model. The generated outputs are then sent to the dynamic risk engine for comprehensive risk assessment.

ZeroTrust AI – Zero Trust Access Monitoring & Analytics System

AI/ML Powered Risk Assessment, RAG Context & Zero Trust Decision

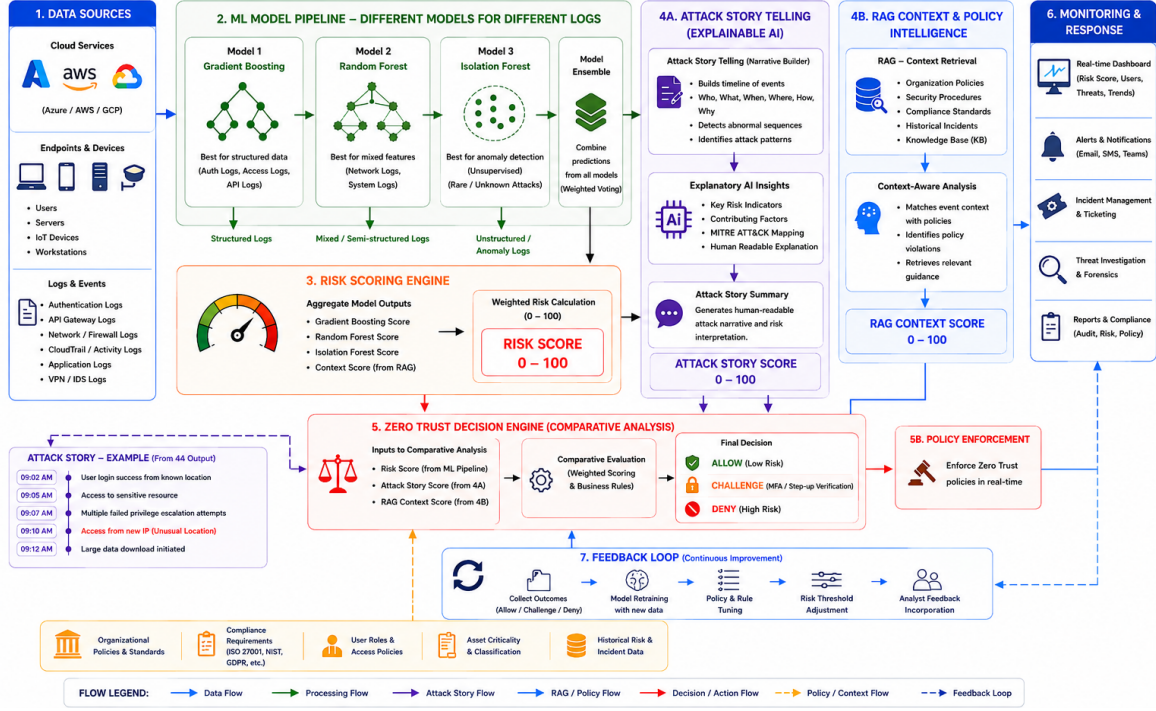


Fig. 1. System architecture of the proposed framework

B. Machine Learning Models and Pipeline

This system uses many ML models, each performing a specific function of security analysis. Random Forest works for classifying known attack types in a supervised manner, and Isolation Forest serves the purpose of anomaly detection for logs and events from the cloud and node. Specific combinations of Gradient Boosting decide how much security risk is present from behavioral characteristics and features. Results from the system's models are treated as additional means of security signal processing and are sent to the risk processing unit instead of being used separately to grant access.

C. Dynamic Risk Scoring and Severity Classification

To operationalize continuous risk assessment within the proposed Zero Trust AI framework, a dynamic risk-scoring mechanism is employed that integrates machine-learning-based threat inference with contextual security intelligence. Rather than relying solely on the output of an individual detection model, the proposed approach synthesizes heterogeneous behavioral and contextual signals into a unified 0–100 risk score.

The final risk score is formulated as:

$$R_{\text{final}} = 0.20R_{\text{ML}} + 0.80R_{\text{Context}}$$

where R_{ML} represents the normalized machine-learning risk score and R_{Context} represents the contextual security risk derived from the characteristics of the access event.

1) Machine-Learning Risk Aggregation: When multiple trained detection models are applicable to an event, their individual risk contributions are aggregated through a weighted normalization mechanism:

$$R_{\text{ML}} = \frac{\sum_{i=1}^n w_i R_i}{\sum_{i=1}^n w_i}$$

where R_i denotes the risk score generated by model i , and w_i denotes its corresponding model weight. The framework assigns weights according to the configured contribution of each detection model:

$$\begin{aligned} w_{\text{NetworkRF}} &= 0.25 \\ w_{\text{AWSIF}} &= 0.20 \\ w_{\text{ZTGB}} &= 0.30 \\ w_{\text{ZTIF}} &= 0.25 \end{aligned}$$

Only models that successfully generate a prediction for the corresponding event contribute to the aggregated ML score, thereby preventing unavailable models from artificially reducing the resulting risk.

2) Context-Aware Risk Computation: The contextual component captures security-relevant characteristics that may not be completely represented by the ML prediction. These include

authentication failures, repeated login attempts, IP reputation, unusual access time, device trustworthiness, sensitive actions and resources, privileged roles, network volume, encryption status, and session duration.

The contextual risk is bounded to the operational range:

$$R_{\text{Context}} = \min \left(100, \sum_{j=1}^m C_j \right)$$

where C_j represents the contribution of an individual contextual security indicator.

This design enables the system to distinguish between events that may receive similar ML predictions but exhibit substantially different operational security contexts.

3) *Risk Severity Classification*: The resulting unified risk score is mapped into four operational severity levels:

$$\text{Severity}(R) = \begin{cases} \text{LOW}, & 0 \leq R < 25 \\ \text{MEDIUM}, & 25 \leq R < 40 \\ \text{HIGH}, & 40 \leq R < 75 \\ \text{CRITICAL}, & 75 \leq R \leq 100 \end{cases}$$

Consequently, the framework transforms heterogeneous access telemetry and ML predictions into an interpretable security decision that can be consumed by monitoring dashboards, alerting mechanisms, and subsequent access-control workflows.

The proposed risk engine therefore establishes a context-aware, model-agnostic risk quantification layer that bridges probabilistic machine-learning inference and operational Zero Trust decision-making, enabling continuous and interpretable assessment of access risk rather than relying on isolated binary anomaly predictions.

D. RAG-Based Policy Analysis

Through the use of the RAG component, the platform has the possibility of reviewing security incidents by utilizing internal policies and regulations. To achieve this objective, the policies are converted into a machine-readable format, segmented into different fragments for easier storage, and included in a knowledge base searchable by using vector information. When a security incident occurs and requires verification of policy compliance, the platform is able to search for relevant information and use it as the background for its examination. Among the advantages of the presented approach, one of the most significant is that there is no need for retraining of models in case of policy revision.

E. Attack Story Reconstruction and Explainable AI

The purpose of an attack story is to link similar incidents together and create a coherent sequence of events. Through time of occurrence, username, session, and sequence information, administrators can track the development of any

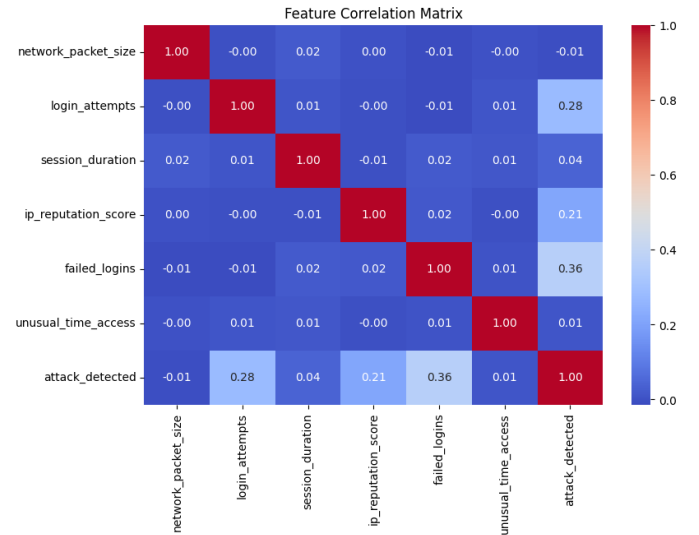


Fig. 2. Feature correlation matrix of security attributes used for attack detection

attacks. Explainable AI offers another level of transparency by identifying the various components that make up a machine learning prediction. Rather than providing just a risk level, the system will also indicate if it was caused by abnormal activity on the network, unusual cloud behaviour, etc.

V. RESULTS AND DISCUSSIONS

A. Experimental Results

Experiments were conducted to evaluate the ability of the proposed ZeroTrustAI framework to detect anomalous activities and quantify security risk under different access conditions. The evaluation considers multiple security attributes, including packet size, login frequency, session duration, IP reputation score, failed login attempts, and access time. The correlation analysis shown in Fig. 2 indicates that attack-related activity is influenced by multiple security attributes rather than a single feature. Among the analyzed attributes, failed login attempts exhibit the highest correlation with attack detection (0.36), followed by login frequency (0.284) and IP reputation score (0.218). The relatively low individual correlation values indicate that multiple security signals should be considered jointly for comprehensive risk assessment.

End-to-end tests were subsequently conducted using representative access scenarios to evaluate the behavior of the proposed risk engine under different security conditions. The results are presented in Table I. The ML-based risk scores remain within a relatively narrow range, while contextual security conditions substantially influence the final risk score. This demonstrates the role of contextual information in differentiating access requests that may have similar machine-learning risk estimates.

The results demonstrate that the centralized risk engine combines security signals from multiple analytical components, including network-based predictions, cloud security analysis, Gradient Boosting risk estimation, and Isolation

TABLE I
ZEROTrustAI END-TO-END RISK TEST RESULTS

#	Scenario	ML Score	Context	Final Score	Severity
1	Normal Employee Access	66.53	10.00	21.31	Low
2	Multiple Failed Logins	64.98	44.50	48.60	High
3	Late Night Financial Access	69.97	36.50	43.19	High
4	Unknown Device Download	71.65	71.50	71.53	High
5	Admin Sensitive Delete	67.50	33.50	40.36	High
6	Highly Suspicious Network	72.75	100.00	94.55	Critical
7	Normal Service Account	64.14	1.00	13.63	Low
8	Possible Data Exfiltration	71.00	86.00	83.00	Critical
9	Normal Manager Activity	67.98	1.00	14.40	Low
10	Highly Suspicious Login	66.74	100.00	93.35	Critical

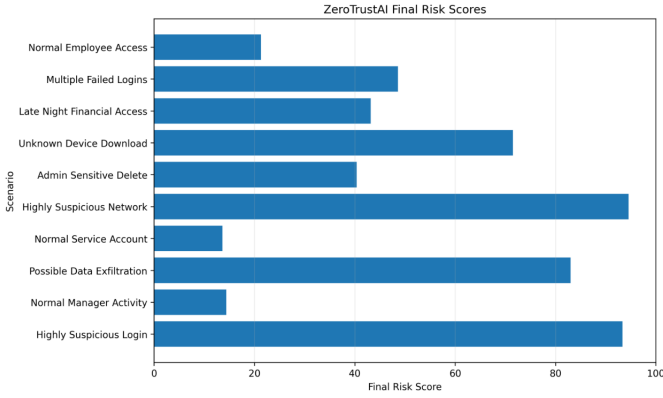


Fig. 3. Risk score variation across different security scenarios

Forest anomaly detection. These outputs are integrated with contextual security information to generate a unified risk score. The contextual component therefore has a significant influence on the final risk assessment, allowing the framework to distinguish between normal activities and scenarios involving suspicious devices, abnormal access patterns, sensitive operations, or potentially malicious network behavior.

As illustrated in Fig. 3, normal activities such as normal employee access, normal service-account activity, and normal manager activity produce relatively low final risk scores of 21.31, 13.63, and 14.40, respectively. In contrast, scenarios involving possible data exfiltration, highly suspicious network activity, and highly suspicious login behavior produce substantially higher scores of 83.00, 94.55, and 93.35, respectively, and are classified as Critical. These results demonstrate that the proposed framework represents security posture as a continuously varying risk state rather than as a simple binary attack or non-attack outcome. Such continuous risk assessment supports Zero Trust access control by allowing security decisions to adapt according to changes in user behavior, device conditions, contextual attributes, and other security signals.

VI. LIMITATIONS AND FUTURE SCOPE

The limitations of the current project arise from its dependence on existing datasets, application logs, and organizationally controlled safety scenarios, which cannot entirely exhibit the degree of complexity and size of big enterprises. The success of machine learning solutions depends on the accuracy and amount of security data accessible; working with different types of data sources as well as processing a large number of events is difficult. Future studies will focus on converting the suggested system into a Low-Code Zero Trust Security Platform, which will allow users to connect various sources of security data, cloud services, applications, and databases with little programming drag. The developed platform gives visual workflows for setting up ML models, security policies, risk-scoring processes, and automated Zero Trust actions, making the system more flexible, scalable, and easier to implement in real organizations.

VII. CONCLUSION

The proposed study shows a Machine Learning-based Zero Trust security framework that integrates Zero Trust Architecture with Machine Learning, RAG, Explainable AI, and continuous security monitoring to resolve the issues found in conventional perimeter-based security. Random Forest, Isolation Forest, and Gradient Boosting perform network threat classification, cloud anomaly detection, and risk evaluation, respectively. With RAG, access activities can be compared with applicable and updatable security policies, while XAI enhances transparency of the decisions made by the ML-based security framework.

The framework integrates the security signals via a single Zero Trust decision mechanism that makes a real-time classification of requests as ALLOW, VERIFY, RESTRICT, or DENY. Continuous monitoring makes it possible for the system to review access once the conditions of the user, device, or activity change. A controlled VirtualBox environment is a safe environment in which normal, anomalous, suspicious,

and policy-violating situations can be tested. In a nutshell, the proposed framework should be a solution that would allow providing adaptive, policy-aware, explainable, and context-sensitive access control, thus boosting the security of modern cloud and distributed environments.

STATEMENTS AND DECLARATIONS

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No funding was received for conducting this study.

Data Availability

This research makes use of the publicly available datasets on cybersecurity and simulated logs of cloud activities. The data processing and analysis techniques described in this paper are reproducible by anyone with access to appropriate publicly available datasets and the implementation information described in the paper. No personally identifiable or confidential organizational data were used in this study.

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